Project Design Phase-II
Technology Stack (Architecture & Stack)

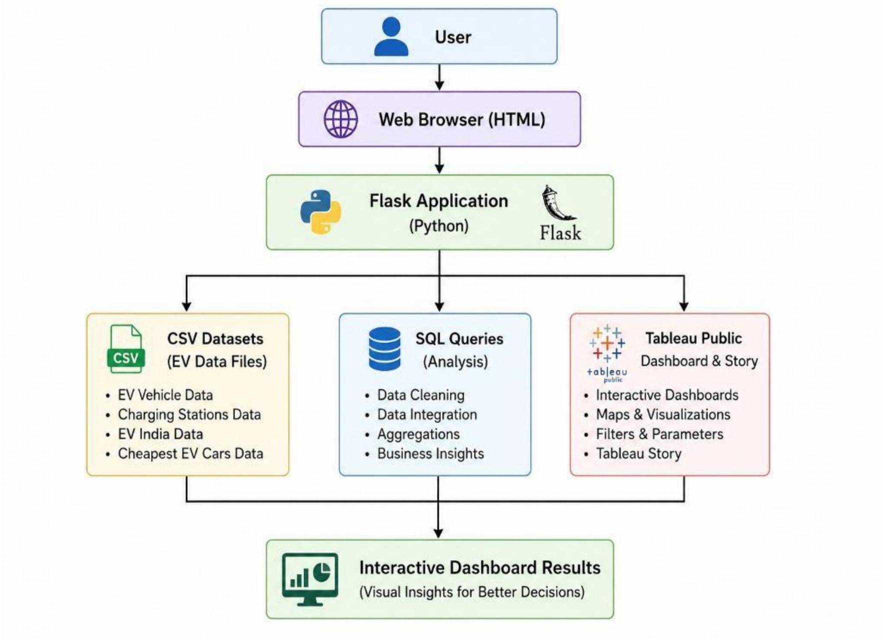
Date	25 June 2026
Team ID	SWTID-2026-4219
Project Name	Visualization Tool for Electric Vehicle Charge and Range Analysis
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/



Guidelines :

1. **Application Logic:** The system uses HTML, CSS, JavaScript, Python (Flask), SQL, and Tableau Public to process EV datasets and generate interactive dashboards and stories.
2. **Infrastructure:** Development is carried out in a **local environment** (VS Code, Flask, SQL, CSV datasets), while deployment and visualization are hosted on **Tableau Public** and **GitHub**.
3. **External Interfaces:** The application integrates with **Tableau Public** for dashboard and story embedding and uses **GitHub** for project hosting and documentation.
4. **Data Storage:** Electric Vehicle datasets, charging station data, SQL scripts, and Tableau workbooks are stored locally in CSV and project files for analysis and visualization.
5. **Machine Learning Interface:** Machine Learning is **not implemented** in the current project but the architecture supports future integration for

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Interactive web application	HTML5, CSS3, JavaScript
2	Application Logic-1	Flask application to host dashboard	Python, Flask
3	Application Logic-2	Dashboard visualization	Tableau Public
4	Application Logic-3	Data preprocessing and integration	Python, Pandas
5	Database	Structured EV datasets	CSV Files
6	Data Source	EV vehicle and charging station datasets	CSV Dataset
7	File Storage	Project files and datasets	Local File System
8	External API-1	Tableau Dashboard Embedding	Tableau Public
9	External API-2	Dashboard & Story Sharing	Tableau Public Share Link
10	Machine Learning Model	Not Applicable	N/A
11	Infrastructure	Local development and GitHub hosting	Local System + GitHub

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open Source Frameworks	Frameworks used in project	Flask, Bootstrap, Tableau Public
2	Security Implementations	Secure embedding through Tableau Public and GitHub repository	HTTPS
3	Scalable Architecture	Modular folder structure separating datasets, SQL, Tableau, screenshots, and documentation	Three-layer architecture
4	Availability	Dashboard accessible online 24×7 through Tableau Public and project available on GitHub	Tableau Public, GitHub
5	Performance	Lightweight datasets with interactive Tableau visualizations for fast rendering	Tableau Public, Flask

References:

https://c4model.com/

https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d